

Common DevOps Syllabus (Proposed for B.Tech - IT & CSE-Cyber Security)

Course Objectives:

- Understand the principles, architecture, and lifecycle of DevOps along with the agile relationship between software development and IT operations.
- Apply version control, source code management, and automation tools (Git, Jenkins, Docker, Kubernetes) for implementing CI/CD pipelines.
- Analyze the effectiveness of containerization, configuration management, and infrastructure-as-code tools (Ansible, Terraform, Puppet, Chef) in optimizing deployment workflows.
- Evaluate software delivery, monitoring, and security practices using DevOps tools such as SonarQube, Prometheus, and ELK Stack for performance and reliability.
- Create integrated DevOps solutions through automation, continuous delivery, and container orchestration to improve collaboration and ensure faster, reliable software releases.

Syllabus :

UNIT I: Introduction to DevOps

Introduction to SDLC, Agile Model, and DevOps Principles. DevOps Architecture, Lifecycle, and Features. Introduction to DevOps Tools: Git, Jenkins, Docker, Kubernetes, Ansible, Terraform. Build Automation, Delivery Automation, Understanding Workflow, Code Quality, and Bottlenecks. Release Management, Scrum, Kanban, and CI/CD Concepts.

UNIT II: Source Code Management (GIT)

Need for Source Code Control and Version Management. History of Git, Roles and Code, Installation and Configuration. Git Workflow, Commands, Branching, Staging, and Collaboration. Unit Testing and Code Coverage: JUnit, NUnit, SonarQube. Code Quality Analysis and Best Practices.

UNIT III: Build Automation and Continuous Integration (CI)

Build Automation Concepts and CI Tools. Jenkins: Architecture, Pipelines, Master-Slave Architecture, and Job Management. Pipeline Basics: Freestyle Projects, Scripted and Declarative Pipelines. User Management, Scheduling Builds, and Launching Builds on Slave Nodes. Integration with Testing Tools such as Selenium and JavaScript Testing.

UNIT IV: Continuous Delivery (CD) and Containerization

Importance of Continuous Delivery and Deployment. Docker: Installation, Commands, DockerFile, Containers, Images, and Docker Hub. Container Orchestration with Kubernetes: Pods, Services, Deployments, ConfigMaps. Kubernetes Architecture, Scaling Applications, and Cloud Platforms (GKE, EKS, AKS). Introduction to CI/CD Tools: Jenkins, GitLab CI, CircleCI.

UNIT V: Configuration Management & Monitoring

Configuration Management Tools: Ansible, Puppet, Chef, Terraform. Infrastructure as Code (IaC): Definition, Benefits, and Tools. Ansible: Tasks, Roles, Jinja Templating, Vaults, and Deployments. DevOps Monitoring Tools: Nagios, Prometheus, Grafana. Log Management: ELK Stack (Elasticsearch, Logstash, Kibana). Security Tools: SonarQube and Secure Deployment Practices.

Course Outcomes:

- **Understand** DevOps principles, architecture, and the integration of agile methodologies in CI/CD workflows.
- **Apply** version control and source code management using Git along with code quality analysis tools such as JUnit and SonarQube.
- **Develop** and **implement** build automation and continuous integration pipelines using Jenkins, Docker, and automated testing with Selenium.
- **Evaluate** configuration management and container orchestration using Ansible, Kubernetes, Puppet, and Chef to enhance deployment efficiency.
- **Analyze** and **monitor** system performance and security using ELK Stack, Grafana, and Prometheus for continuous improvement and reliability.

Text Books:

1. Joyner, Joseph., Devops for Beginners: Devops Software Development Method Guide for Software Developers and It Professionals, 1st Edition Mihails Konoplow, 2015.
2. Alisson Machado de Menezes., Hands-on DevOps with Linux,1st Edition, BPB Publications, India, 2021.
3. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations Paperback – October 6, 2016, 1st Edition.
4. Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation (Addison-Wesley Signature Series (Fowler)) 1st Edition by Jez Humble (Author), David Farley (Author), 1st Edition, 2010.
5. Docker Deep Dive Paperback – July 12, 2017 by Nigel Poulton (Author), 1st Edition.
6. Terraform: Up & Running, 2nd Edition by Yevgeniy Brikman Released September 2019 Publisher(s): O'Reilly Media, Inc.
7. The entire Pro Git book, written by Scott Chacon and Ben Straub and published by Apress, first edition, 2014.
8. What is Devops? Infrastructure as code, 1st Edition, Mike Loukides ,O'Reilly publications, 2012.

Reference Books:

1. Software Engineering: A Practitioner's Approach" – Roger S. Pressman
2. Building a DevOps Culture, 1st Edition, Mandi Walls, O'Reilly publications,2013.
3. The DevOps 2.0 Toolkit: Automating the Continuous Deployment Pipeline With Containerized Micro services, 1st Edition, Viktor Farcic, CreateSpace Independent Publishing Platform publications,2016
4. Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation, 1st Edition, Jez Humble and David Farley,2010.
5. Achieving DevOps: A Novel About Delivering the Best of Agile,DevOps

Web References:

https://www.tutorialspoint.com/sdlc/sdlc_overview.htm

<https://www.agilealliance.org/agile101/>

<https://scrumguides.org/>

<https://aws.amazon.com/devops/what-is-devops/>

[What is DevOps? - DevOps Models Explained - Amazon Web Services \(AWS\)](#)

[SDLC Overview](#)

[DevOps Roadmap: Learn to become a DevOps Engineer or SRE](#)

[Complete DevOps Roadmap 2025 - Master these 4 Levels!](#)

[Kunal Kushwaha - YouTube](#)

<https://www.sonarsource.com/products/sonarqube/>